

Programming Lab 4 — Loops

Case Problem Labs

In this lab, you will continue working with your assigned Case Problem business. *You can find your assigned case problem group by clicking the Groups icon in the main Canvas navigation pane.*

You will create a Python program that uses loops to process repeated data, track totals, and generate summary information.

This Lab Reinforces

- while loops
- sentinel-controlled loops
- counters
- accumulators
- running totals
- averages
- basic branching inside loops
- formatted summaries
- real-world business processing

Important Note

Programming labs often require multiple attempts and revisions. Debugging and refining your program are normal parts of learning to program.

★ Challenge Extension (Optional)

Challenge Extensions are optional enhancements for students who would like additional practice or a more advanced programming challenge. Students should focus on completing the required functionality first before attempting extensions.

[Fountain View Hall](#)

[Cost Is No Object](#)

[The Barking Lot](#)

Fountain View Hall

Lab 4: Event Attendance & Revenue Tracker

Overview

Fountain View Hall wants a program that tracks event attendance and projected revenue for multiple bookings.

Program Requirements

- Repeatedly prompt for event name, guest count, and estimated revenue
 - Continue until the user enters "done"
 - Track total events processed
 - Track total guests
 - Track total projected revenue
 - Calculate the average guests per event
 - Determine the largest event
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Business Rules

- **Pricing Rules**
 - Large events receive a discount
- **Operational Rules**
 - Every booking must include a guest count
 - Booking total must be calculated before the discount is applied
 - Revenue totals should be tracked

Assumptions & Pricing

- Not all pricing values and business rules are fully defined. You may make reasonable assumptions for rates, fees, discounts, and surcharges as needed to complete your program. Clearly document any assumptions in your code using comments.
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Program Must

- Use a while loop
 - Use counters and accumulators
 - Use a sentinel value to end the loop
 - Display a formatted summary report
 - Include at least 3 comments
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Example Summary Output

Event Revenue Summary

Events Processed: 6

Total Guests: 1,245

Total Revenue: \$18,450.00

Average Guests Per Event: 207.5

Largest Event: 420 Guests

★ Challenge Extension (Optional)

Add:

- Weekend event counter
- Large event warning messages
- Premium catering counter

Submission Requirements

Submit:

- .py file
 - Test cases
 - Code walk-through video (3–4 minutes)
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Cost Is No Object

Lab 4: Daily Rental Summary System

Overview

Cost Is Not Object wants a program that tracks rental activity throughout the day and generates a summary report of rental transactions.

Program Requirements

- Repeatedly prompt for customer name, rental amount, and vehicle type.
- Continue processing until the user enters "done"
- Track the total number of rentals processed
- Track total rental revenue
- Calculate the average rental amount
- Determine the highest rental amount

Business Rules

- **Pricing Rules**
 - Luxury vehicles include a surcharge
- **Operational Rules**
 - Rental total must be calculated before the discount is applied
 - Revenue totals should be tracked

Assumptions & Pricing

- Not all pricing values and business rules are fully defined. You may make reasonable assumptions for rates, fees, discounts, and surcharges as needed to complete your program. Clearly document any assumptions in your code using comments.

Program Must

- Use a while loop
- Use counters and accumulators
- Use a sentinel value to end the loop
- Display a formatted summary report
- Include at least 3 comments

Example Summary Output

Daily Rental Summary

Rentals Processed: 8

Total Luxury Vehicles: 3

Total Revenue: \$4,820.50

Average Rental Amount: \$602.56

Highest Rental Amount: \$1,250.00

★ Challenge Extension (Optional)

Add:

- Luxury rental counter
- Premium membership counter
- Discount eligibility messages

Submission Requirements

Submit:

- .py file
 - Test cases
 - Code walk-through video (3–4 minutes)
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The Barking Lot

Lab 4: Kennel Occupancy Tracker

Overview

The Barking Lot wants a program that tracks kennel occupancy and boarding statistics throughout the day.

Program Requirements

- Repeatedly prompt for dog name and boarding days
- Continue until the user enters "quit"
- Track total number of dogs checked in
- Track total boarding days for all dogs
- Calculate the average boarding stay length
- Determine the longest stay

Business Rules

- **Pricing Rules**
 - Boarding stays longer than 10 days receive a discount
- **Operational Rules**
 - Each dog must have a boarding duration
 - Boarding stay total must be calculated before the discount is applied

Assumptions & Pricing

- Not all pricing values and business rules are fully defined. You may make reasonable assumptions for rates, fees, discounts, and surcharges as needed to complete your program. Clearly document any assumptions in your code using comments.

Program Must

- Use a while loop
- Use counters and accumulators
- Use a sentinel value to end the loop
- Display a formatted summary report
- Include at least 3 comments

Example Summary Output

Kennel Occupancy Summary

Dogs Checked In: 12

Total Boarding Revenue: \$2,754.50

Total Boarding Days: 54

Average Stay Length: 4.5 days

Longest Stay: 11 days

★ **Challenge Extension (Optional)**

Add:

- VIP package counter
- Discount eligibility messages
- Manager approval notifications

Submission Requirements

Submit:

- .py file
 - Test cases
 - Code walk-through video (3–4 minutes)
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